## DTC P0471:00 [SKYACTIV-D 1.5]

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| DTC P0471:00 | Exhaust gas pressure sensor No.1 circuit range/performance problem |
|---|---|
| DETECTION CONDITION | • When any of the following conditions is met:<br>— With the following conditions met, a condition has continued for **1.8 s** within **4 to 6 s** after the ignition was switched OFF in which difference between the barometric pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm$^2$, 7.30 psi} or more** and difference between the intake air pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm$^2$, 7.30 psi} or more**.<br>• Battery voltage: **10 V or more**<br>— The PCM detects a condition two times in which difference between the maximum exhaust gas pressure when the following condition A is met and the minimum exhaust gas pressure when the following condition B is met is less than **11 kPa {0.11 kgf/cm$^2$, 1.6 psi}**.<br>**Condition A: the following conditions continue for 0.32 s or more**<br>• Battery voltage: **8—20 V**<br>• **0.5 s** have elapsed after the ignition was switched ON (engine off or on)<br>• Engine coolant temperature: **55 °C {131 °F} or more**<br>• Intake air temperature: **−10 °C {14 °F} or more**<br>• Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm$^2$, 9.5—17.4 psi}**<br>• Change in fuel injection amount: **Less than 4 mm$^3$/st**<br>• Vehicle speed: **above 60 km/h {37 mph}**<br>• Target intake air pressure is **150 kPa {1.53 kgf/cm$^2$, 21.8 psi} or more** for **1.5 s or more**<br>• Difference in fuel injection amount before and after correction is **less than 5 mm$^3$/st**<br>• VSS malfunction DTC P0500:00 not detected<br>• Exhaust gas pressure sensor No.1 malfunction DTC P0472:00, P0473:00 not detected<br>• Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected<br>• Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected<br>• Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected<br>**Condition B: the following conditions continue for 0.32 s or more**<br>• Battery voltage: **8—20 V**<br>• **0.5 s** have elapsed after the ignition was switched ON (engine off or on)<br>• Engine coolant temperature: **40 °C {104 °F} or more**<br>• Intake air temperature: **−7 °C {19 °F} or more**<br>• Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm$^2$, 9.5—17.4 psi}**<br>• Change in fuel injection amount: **Less than 4 mm$^3$/st** within **0.016 s**<br>• Fuel injection amount learning is continued for **2 s or more**<br>• VSS malfunction DTC P0500:00 not detected<br>• Exhaust gas pressure sensor No.1 malfunction DTC P0472:00, P0473:00 not detected<br>• Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected<br>• Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected<br>• Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected<br>**Diagnostic support note**<br>• This is an intermittent monitor (turbo system).<br>• The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM.<br>• PENDING CODE is available if the PCM detects the above malfunction condition during first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| FAIL-SAFE FUNCTION | • PCM restricts engine torque.<br>• Inhibits the EGR control.<br>• PCM restricts engine-transaxle integration control. (ATX) |
| POSSIBLE CAUSE | • Exhaust gas pressure sensor No.1 connector or terminals malfunction<br>• Exhaust gas pressure sensor No.1 malfunction<br>• PCM connector or terminals malfunction<br>• PCM malfunction |
| SYSTEM WIRING DIAGRAM | Not applicable |

**Diagnostic Procedure**

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION**<br><br>**Note**<br>• Recording can be facilitated using the screen capture function of the PC.<br><br>• Record the FREEZE FRAME DATA/snapshot data on the repair order. | — | Go to the next step. |
| 2 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 3 | **INSPECT EXHAUST GAS PRESSURE SENSOR NO.1 CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the exhaust gas pressure sensor No. 1 connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 6. |
| | | No | Go to the next step. |
| 4 | **INSPECT EXHAUST GAS PRESSURE SENSOR NO.1**<br>• Reconnect all disconnected connectors.<br>• Inspect the exhaust gas pressure sensor No.1. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the exhaust gas pressure sensor No.1, then go to Step 6.<br>(See EXHAUST GAS PRESSURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 5 | **INSPECT PCM CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to the next step. |
| | | No | Go to the next step. |
| 6 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Start the engine and idle it.<br>• Switch the ignition off and wait for **10 s or more**.<br>• Drive the vehicle on a flat road surface **10 times** under the following conditions.<br>1. Idle the engine for **5 s**. (engine stop during i-stop operation)<br>2. Accelerate to **40 km/h {25 mph}** within **5 s**.<br>3. Stop the vehicle.<br>• Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the PENDING CODE for this DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 7 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |